TEXORA · LECTURE SHEET

The Apparel Industry

How clothing production grew from the tailor's bench to the modern export factory, the language the industry speaks, the sequence every garment follows, and the fabrics it works with.

TE-2207 · APPAREL MANUFACTURING-1 · TOPIC 1 · LEVEL 2 TERM 2

1. What the apparel industry is

The **apparel industry** — also called the garment industry or, in Bangladesh, the **ready-made garment (RMG)** industry — is the branch of the textile chain that converts fabric into finished clothing. It is the last manufacturing stage before the product reaches the retailer and the wearer, and it is by far the most labour-intensive stage: spinning, weaving and dyeing are done by machines tended by relatively few people, while cutting, sewing and finishing still need many skilled hands at many workstations.

The full textile chain runs: **fibre → yarn → fabric → wet processing (dyeing, printing, finishing) → apparel → distribution and retail**. The apparel factory sits at the conversion point where a flat, two-dimensional material becomes a three-dimensional product shaped to the human body. Its core operations — pattern making, marker making, spreading, cutting and sewing — are the subject of this course, TE-2207, which covers everything up to and including cutting and the preparation for sewing; sewing itself and garment finishing are taken further in the following term.

> **DEFINITION**
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> Apparel manufacturing is the organised conversion of fabric into finished garments through a sequence of preparatory operations (pattern, marker, spreading, cutting) and assembly operations (sewing, finishing, inspection, packing), carried out to a buyer's specification of style, quality, quantity and delivery date.

Factories differ in what they take responsibility for. A **CMT** unit works on *cut, make and trim* terms — the buyer supplies fabric and trims and pays only for the conversion. An **FOB** manufacturer buys the fabric and trims itself and sells the finished garment at a full price. Between the two sit intermediate arrangements, and the trend in Bangladesh has been a steady move from CMT towards FOB, because the manufacturer then controls fabric sourcing and delivery more completely.

2. Historical development — the world

For most of history, clothing was made one piece at a time: at home, or by a tailor working to an individual customer's measurements. Three developments turned that craft into an industry.

- **The sewing machine (1840s–1850s).** Elias Howe patented a practical lockstitch machine in 1846 and Isaac Singer's improved, commercially successful version followed in 1851. A machine-sewn seam could be produced many times faster than a hand-sewn one, and for the first time identical garments could be made in quantity.
- **Standard sizes and ready-made clothing (mid-1800s onwards).** Military uniforms in wartime created demand for clothing in graded size sets rather than individual fits. The same measurement data made civilian ready-to-wear possible: a customer could buy a finished garment off the rail instead of commissioning one.
- **The factory system and power (late 1800s–1900s).** Rows of machines driven first by line shafts and then by individual electric motors, work divided into small operations along a line, and electric cutting knives replaced the tailor's shears. Production engineering — operation breakdown, line balancing, work study — entered the factory in the twentieth century.

Through the first half of the twentieth century the industry was centred in the United States and Western Europe. Rising labour costs then pushed production steadily towards lower-wage economies: first Japan, then Hong Kong, Taiwan and South Korea in the 1960s–70s, then China and South and South-East Asia from the 1980s. China became the world's largest exporter; Bangladesh, Vietnam, India and others followed as large, competitive producers. The pattern is consistent: apparel manufacture migrates towards skill, reliability and low labour cost combined, and stays where productivity, compliance and logistics keep improving.

3. Historical development — Bangladesh

Bangladesh's entry came late but grew very fast. The milestones most often recorded are:

- **1960s:** small beginnings — Reaz Garments, established in Dhaka in 1960, is commonly cited as the country's first garment factory, initially serving the local market and exporting in small volume from the 1970s.
- **1978–79:** the decisive step. Desh Garments, founded by Noorul Quader, entered a technical collaboration with the South Korean company Daewoo, and in 1979 a group of about 130 Bangladeshi workers and managers were trained in Korea. Many of them went on to found or run factories of their own — the trained group seeded an industry.
- **1980s:** rapid expansion under the Multi-Fibre Arrangement (MFA), which guaranteed market access through quotas, together with bonded-warehouse and back-to-back letter-of-credit facilities that let factories import fabric against export orders. Hundreds of factories opened.

- **1990s:** knitwear grew alongside woven garments, child-labour concerns were addressed under international pressure, and backward-linkage spinning, knitting and dyeing capacity developed inside the country.
- **2005:** the MFA quota system ended. Against predictions of collapse, Bangladesh continued to grow on price, capacity and an expanding knitwear base.
- **2013 onwards:** the Rana Plaza collapse forced a structural change in factory safety; the Accord and Alliance programmes inspected and remediated factories, followed by the national RMG Sustainability Council (RSC). Compliance became a condition of doing business, and Bangladesh built a large number of certified green garment factories.

Today RMG is Bangladesh's dominant export sector — the large majority of the country's export earnings — and Bangladesh stands among the world's largest clothing exporters, supplying major buyers in Europe and North America. The industry employs several million workers, the majority of them women from rural areas, which is why its history is also a social history: it changed household incomes, women's employment and urbanisation across the country. The engineer's interest in this history is practical: the factory system, the compliance framework and the buyer's way of working that this course teaches all grew out of these decades.

4. Apparel terms and definitions

The industry has a working vocabulary that a technologist must read fluently. The most-used terms:

TERM	MEANING
Garment / apparel	A finished article of clothing; “apparel” is the collective trade term.
Style	A particular design identified by a style number; one order may contain several styles.
Buyer	The importing company or brand that places the order (retailer, wholesaler or label).
Vendor / manufacturer	The factory or exporting company that makes the garment for the buyer.
Specification (spec) sheet	The document giving measurements, construction, materials and standards for a style (Topic 2).
Tech pack	The full technical package from the buyer: spec sheet, sketches, bill of materials, construction and packing details.
Proto sample	The first sample made to show design intent, before fitting.
Fit sample	A sample checked on a model or dress form for fit and measurement.

PP (pre-production) sample	The sample made with actual bulk fabric and trims before production starts; the approved PP sample is the production standard.
Shipment sample	A sample taken from the finished bulk production and sent to the buyer.
Trims / accessories	Everything on the garment besides the main fabric: buttons, zips, thread, labels, elastic, tags, polybags.
CMT	Cut, Make, Trim — conversion-only terms; buyer supplies materials.
FOB	Free On Board — the price of the finished garment delivered to the port of shipment, materials included.
Lead time	The total time from order confirmation to shipment.
AQL	Acceptable Quality Limit — the statistical sampling standard used in final inspection.

5. The apparel manufacturing sequence

Whatever the garment, production follows the same backbone sequence. The chart shows the full route; this course deals in depth with the steps from pattern to bundle.

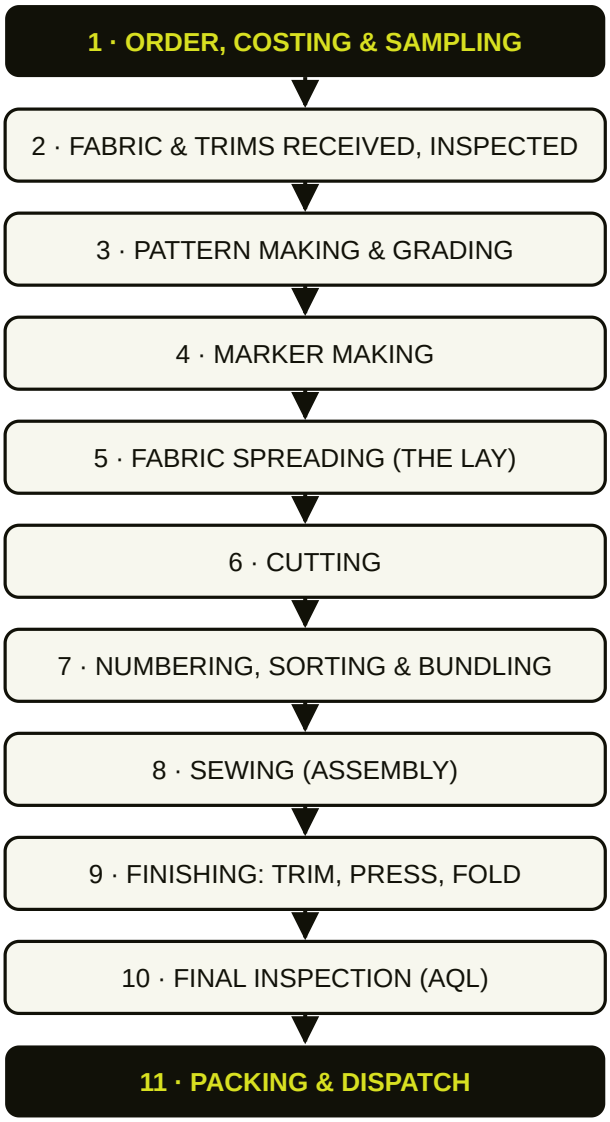


Figure 1 — The apparel manufacturing sequence. Steps 3–7 are the preparatory operations studied in this course.

Three features of the sequence matter. First, it is **sequential with quality gates**: a defect not caught at fabric inspection will be sewn into a garment and cost far more to correct later, so inspection points sit before the expensive steps. Second, the preparatory operations decide the economics: fabric is the largest single cost in most garments, and pattern, marker and cutting decide how much of it becomes clothing and how much becomes waste. Third, the sequence runs in **bulk under a sample's authority** — nothing is cut for production until the PP sample is approved.

6. Sample apparel making

A **sample** is a garment made before bulk production to test and approve design, fit, materials and workmanship. Sampling is where the buyer's idea and the factory's capability are reconciled; mistakes

found in a sample cost almost nothing, while the same mistakes in bulk are catastrophic. The usual progression is:

1. **Proto (prototype) sample** — first interpretation of the design, often in substitute fabric; checks the look and construction idea.
2. **Fit sample** — made to the spec measurements and checked on a live model or standard form; measurements are corrected from this stage.
3. **Size-set sample** — one garment in every size, proving the grading across the size range.
4. **PP (pre-production) sample** — made in the actual bulk fabric and trims, by the actual production methods; the buyer's approval of the PP sample authorises bulk cutting.
5. **Production (top-of-production) sample** — taken from the first garments off the line to confirm bulk matches the PP sample.
6. **Shipment sample** — drawn from packed production and sent as evidence of what was shipped.

Each sample goes through the same mini-sequence as production — pattern, marker, spreading, cutting, sewing, finishing — but one garment at a time, usually in a dedicated sample room with the most experienced machinists. The comments returned by the buyer on each round are recorded, patterns are corrected, and the next sample incorporates them; this loop, not any single sample, is the real sampling process.

7. Fabrics used in the apparel sector

Apparel fabrics divide by construction into **woven**, **knitted** and (a small share) **nonwoven** interlinings and accessories. The choice is made by the garment's function: woven fabrics are stable and hold shape for shirts, trousers and suits; knits stretch and give comfort for T-shirts, polo shirts, underwear and sportswear.

FABRIC (CONSTRUCTION)	TYPICAL APPAREL USES
Poplin / plain weave (woven)	Formal and casual shirts, dresses, blouses
Twill and denim (woven)	Jeans, chinos, workwear, jackets
Gabardine, canvas, corduroy (woven)	Trousers, uniforms, outerwear
Satin, georgette, chiffon (woven)	Ladies' wear, scarves, evening wear
Single jersey (knit)	T-shirts, underwear, casual tops

Rib and interlock (knit)	Cuffs and necklines, underwear, polo bodies
Piqué (knit)	Polo shirts
Fleece and terry (knit)	Winter wear, hoodies, towels, baby wear
Nonwoven interlining	Collars, cuffs, waistbands — fusible support inside garments

The fabric also dictates preparatory handling throughout this course: knits need relaxation before cutting and care against stretching during spreading; checked and striped wovens need matching in the marker and at the sewing line; pile fabrics such as corduroy must be spread in one nap direction. Fabric is never a neutral raw material — each course topic that follows will keep meeting its conditions.

8. Glossary

TERM	MEANING
RMG	Ready-made garments; the standard term for Bangladesh's export garment sector.
Preparatory operations	The operations before sewing: pattern making, marker making, spreading, cutting, sorting and bundling.
Backward linkage	Domestic industries supplying the garment factories — spinning, knitting/weaving, dyeing and trims.
Compliance	Meeting the legal and buyer-required standards of safety, labour and environment in a factory.
Bundle	A tied group of cut panels for one garment part or a set of garments, moving through sewing as a unit.

9. Key points and summary

- The apparel industry converts fabric into garments; it is the final, most labour-intensive stage of the textile chain.
- The sewing machine, standard sizing and the factory system turned tailoring into mass production; production then migrated across the world towards favourable cost and skill.
- Bangladesh entered through Reaz Garments and Desh–Daewoo (late 1970s), grew under MFA quotas, survived the 2005 quota end, and rebuilt around safety compliance after 2013.
- Production follows a fixed backbone — order and sampling, inspection, pattern, marker, spreading, cutting, bundling, sewing, finishing, inspection, packing — and samples, above all the PP sample, authorise every bulk step.

- Fabric type (woven or knitted) decides garment function and conditions every preparatory operation that follows.

10. Practice

Answer in your own words; brief model answers follow.

1. Place the apparel factory in the textile chain and state two reasons why the apparel stage employs far more people per unit of output than spinning or weaving.
2. Why is the Desh–Daewoo episode of 1979 treated as the real beginning of Bangladesh's export garment industry, rather than the earlier factories?
3. A buyer returns comments on a fit sample. Which samples must still be made before bulk cutting, and what exactly does the final one authorise?
4. Arrange into production order: bundling, pattern grading, packing, spreading, final inspection, marker making, sewing.
5. A style uses checked twill for trousers and single jersey for T-shirts. Give one preparatory consequence of each fabric choice.
6. Explain the difference between CMT and FOB terms, and why a factory usually prefers to grow from CMT to FOB.

Answer key

1. Apparel is the last manufacturing stage, converting dyed/finished fabric into garments. It is labour-intensive because garments are limp, three-dimensional assemblies: panels must be guided, aligned and sewn piece by piece, and inspection/finishing are manual; spinning and weaving are continuous machine processes tended by few operators.
2. Because the ~130 trainees who returned from Korea carried both production skills and buyer know-how, and they founded or staffed many factories — the episode multiplied capability instead of remaining a single factory.
3. Size-set samples, then the PP sample in actual fabric and trims. Approval of the PP sample authorises bulk cutting and fixes the standard the bulk must match.
4. Pattern grading → marker making → spreading → bundling (cut panels, after cutting) → sewing → final inspection → packing.
5. Checked twill: patterns must be matched at seams, so the marker and lay must align checks (and the fabric is stable, easing spreading). Single jersey: it curls and stretches, so it must be relaxed before cutting and spread without tension.
6. Under CMT the buyer supplies fabric and trims and pays only conversion charges; under FOB the factory sources materials and sells the finished garment at a full price. FOB gives the factory control

over materials and delivery and a larger share of the value — but also the financing and sourcing risk.
